

Dongchan Kim

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Aug. 2026 – May 2028 (Expected)

M.S. in Systems and Entrepreneurial Engineering

Champaign, IL

Incoming Research Assistant, HXRI Lab – Advisor: Prof. Avinash Gupta; fully funded with tuition waiver.

Stony Brook University (SUNY)

Graduated Jun. 2026

B.S. in Technology Systems Management, Computer Science Specialization; GPA: 3.77/4.00; Magna Cum Laude

Stony Brook, NY

Coursework: Data Structures, Object-Oriented Programming, Computer Networks, Database Design, Scripting Languages.

PROFESSIONAL EXPERIENCE

KAI Inc. — Studio Realive Project for SM Entertainment

Jun. 2025 – Sep. 2025

Software Engineer Intern – Commercial Multi-HMD VR Theater

Daejeon, South Korea

- Designed and operated a Socket.IO synchronization service for 8K VR playback, applying round-trip-time correction and WebSockets room state to achieve +/-10 ms trigger alignment and keep frame variance below 50 ms across concurrent client headsets.
- Refactored a 3,000-line monolithic controller into 4 testable modules using dependency injection and single-responsibility design, diagnosed and resolved persistent memory leaks observed during multi-headset sessions.
- Collaborated with an external CMS team to integrate Unity and Node.js services, then automated testing and containerized deployments through GitHub Actions and Docker, replacing the previous manual release workflow.

TeamSparta Inc.

Jul. 2024 – Aug. 2025

Technical Interview Tutor

Remote

- Mentored 500+ developers and reviewed 1,000+ Python, JavaScript, and SQL submissions covering data structures and algorithms; students achieved a 25% higher assessment pass rate and 30% faster average solution execution.

PROJECTS

SmartSight – MIT Reality Hack Gold Prize & Meta Track Winner ([Devpost](#) | [Meta Blog](#))

Node.js, PostgreSQL, AWS

- Architected the backend for SmartSight, a hands-free wearable learning platform, with 47+ REST endpoints, 14 Prisma models, WebSockets communication, and multi-model orchestration.
- Reduced end-to-end response latency by 54% (13s to 6s) using asynchronous event pipelines and reduced database queries by 50x through tool-level query consolidation.

A.P.E – AWS Platform Explorer ([GitHub](#))

Go, TypeScript, AWS

- Built and released a 16 MB cross-platform AWS operations tool by embedding a React/TypeScript UI within a single Go binary for browsing EC2 files, S3 objects, and RDS PostgreSQL databases.
- Implemented an in-process SSH tunnel through an EC2 bastion for private-subnet RDS access; kept credentials local and added database/schema drill-down with generated ERDs.
- Automated cross-platform releases with CI, GoReleaser, and Homebrew, and collaborated with external contributors to ship community requested improvements.

Tryl – AI Virtual Try-On Platform ([Home](#) | [Chrome Web Store](#))

Python, Prometheus, Grafana, Redis, AWS

- Founded and launched an end-to-end AI virtual try-on platform spanning a Manifest Chrome extension, Next.js web app, FastAPI backend, PostgreSQL, and an asynchronous Python AI worker.
- Implemented a Redis job queue with retry handling and temporary caching, allowing long-running AI generation tasks to run separately from API requests.
- Automated testing and EC2 deployment with GitHub Actions and Docker, running 19 API tests before building three service images; integrated Prometheus and Grafana to collect and visualize service performance metrics.

HONORS & LEADERSHIP

- MIT Reality Hack 2026** – Gold Prize and Meta Track Winner for SmartSight, a hands-free Smart Glasses learning platform.
- SeSAC Hackathon 2024 (Google & Seoul Job Festival)** – Honorable Mention (Top 10 Team); Team Lead.
- Lambda, Side Project Club** – Vice President and Code2Career Lead; built a developer-learning community that increased applications by 350% year over year and later returned as an invited alumni speaker.
- Academic Honors** – Dean's List (four semesters); Achievement and Encouragement Scholarships (\$3,000 total).

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, Bash, Go

Backend, Systems & Data: Node.js, FastAPI, REST APIs, WebSockets, PostgreSQL, Redis

Cloud, DevOps & Observability: AWS (EC2, RDS, S3, CloudWatch), Docker, GitHub Actions, CI/CD, Unix/Linux, Prometheus, Grafana

AI/ML: OpenCV, NumPy, Gemini/OpenAI APIs

Spoken Languages: English (Fluent), Korean (Native)